

MODULE 4

TON, TOF AND TP TIMERS

Siemens TIA Portal PLC Course

Module 4: TON, TOF AND TP TIMERS

Author: AUTOMATION AND ROBOTICS ENGINEER

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Introduction

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This module covers the operation and application of industrial timers in automation systems programmed with Siemens PLCs. Timers are fundamental tools for controlling sequences, delays, waiting times, and activation pulses within an industrial process.

In the Siemens TIA Portal V20 programming environment, the TON, TOF, and TP timer blocks make it possible to develop time-based control logic, facilitating the automation of processes such as sequential startups, lighting systems, motor control, pneumatic actuators, pumping systems, and production processes.

The goal of this module is for the student to understand the operating principle of each timer, know its main parameters, and be able to implement them correctly in Ladder (LAD) language to solve real industrial applications.

The Concept of Timing in PLC Systems

A timer is a logic instruction that generates a response after a set interval of time. Unlike a digital input, whose activation depends directly on a field signal, a timer allows the PLC to make decisions based on elapsed time.

For example, an input signal can start a 5-second timer. During that period, the PLC counts the programmed time and, once the set value is reached, changes the state of its logic output.

Timers are widely used to:

- Generate activation delays.
- Keep an output active for a set period.
- Generate pulses of a defined duration.
- Coordinate operating sequences.
- Control waiting times between process stages.
- Prevent simultaneous equipment startups.
- Control lighting and ventilation systems.
- Manage actuators and pneumatic processes.

Types of Timers in TIA Portal

The main timers used in this module are:

- TON: On-delay timer.
- TOF: Off-delay timer.
- TP: Pulse-generation timer.

Each one behaves differently and must be selected according to the specific requirements of the automated process.

TON Timer — On-Delay

The TON timer, known as Timer On Delay, delays the activation of its output for a set period.

When the enable input IN goes to logic TRUE, the timer starts counting the time set in the PT parameter. The output Q remains FALSE during the timing period. Once the elapsed time reaches the preset value, output Q switches to TRUE.

TON Operation

- Input IN goes to logic TRUE.
- The timer starts counting time.
- Output Q remains off.
- Elapsed time ET increases progressively.
- When ET reaches the time programmed in PT, output Q turns on.
- If input IN returns to FALSE, the timer resets.

Main parameters

Parameter	Description
IN	Timer enable input.
PT	Preset time or setpoint time.
Q	Timer output.
ET	Elapsed time.

Industrial example

A motor must start five seconds after a run signal is activated. The run signal enables the TON timer. During the first five seconds, the timer output remains off. Once the programmed time has elapsed, output Q can energize the motor contactor.

This type of application is common in:

- Sequential motor starting.
- Pump activation delay.
- Ventilation systems.
- Heating processes.
- Conveying systems.

TOF Timer — Off-Delay

The TOF timer, known as Timer Off Delay, keeps an output active for a period of time after the input signal has been deactivated.

While input IN remains TRUE, output Q stays active. When the input goes to FALSE, the timer starts counting the set time. During this period, output Q remains active. Once the programmed time is reached, output Q switches to FALSE.

TOF Operation

- Input IN is activated.
- Output Q is activated immediately.

- Input IN goes to FALSE.
- The timer starts counting the off-delay time.
- Output Q remains active during the programmed time.
- When the time ends, output Q turns off.

Industrial applications

The TOF timer is mainly used in applications where an output must remain active for an additional period after the command signal disappears. Some applications are:

- Fans that must keep running after a process stops.
- Air extraction systems.
- Timed lighting.
- Motor cooling.
- Lubrication systems.
- Off-delay of actuators.

Practical example

An extraction fan runs while a main motor is active. When the motor stops, the fan must keep running for 30 seconds to dissipate accumulated heat.

In this case, the main motor activates the TOF timer's IN input. When the motor stops, the timer keeps the output active for the programmed 30 seconds before switching off the fan.

TP Timer — Pulse Generator

The TP timer, known as Pulse Timer, generates an output pulse of a set duration.

When input IN receives a trigger, output Q switches to TRUE and stays active for the time set in PT. Once the programmed time ends, the output turns off automatically. The TP timer is especially useful when an actuator needs to be activated for a specific time.

TP Operation

- Input IN receives a trigger signal.
- Output Q is activated.
- Counting of the set time begins.
- The output remains active during the programmed time.
- When the time ends, output Q turns off automatically.

Industrial applications

- Activating a solenoid valve for a set time.
- Generating command pulses.
- Controlling pneumatic cylinders.
- Triggering alarms.
- Controlling dosing systems.
- Driving ejection mechanisms.
- Generating timed signals for other processes.

Practical example

A pneumatic cylinder must extend for three seconds to eject a defective part from a production line. Detecting the part activates the TP timer's IN input. Output Q energizes the solenoid valve for three seconds and then switches off automatically.

Comparison Between TON, TOF, and TP

Timer	Main function	Behavior
TON	On-delay	Activates the output after a delay
TOF	Off-delay	Keeps the output active for a set time
TP	Pulse generation	Activates the output for a fixed time

Choosing the right timer depends on the operating sequence to be implemented.

Input/Output Factor

The following addressing scheme will be used for the module's practical exercise:

Element	Type	Address	Description
Start pushbutton	Digital input	I0.0	Starts the process
Stop pushbutton	Digital input	I0.1	Stops the process
Presence sensor	Digital input	I0.2	Detects the part
Process sensor	Digital input	I0.3	Confirms process condition
Motor	Digital output	Q0.0	Drives the motor
Solenoid valve	Digital output	Q0.1	Controls the pneumatic actuator
Run indicator	Digital output	Q0.2	Indicates process is active
Fault indicator	Digital output	Q0.3	Indicates a fault condition

The I/O Factor relates the physical process signals to the addresses used within the PLC program. Correctly defining it makes programming, commissioning, and later maintenance of the system easier.

Process Operating Sequence

The automated process runs as follows:

- The operator presses the Start pushbutton I0.0.
- The system starts the process and activates motor Q0.0.
- Run indicator Q0.2 shows that the system is operating.
- When sensor I0.2 detects a part, the corresponding timer starts.
- Depending on the programmed sequence, an activation delay is generated via TON.
- Once the programmed time elapses, the next process stage is activated.
- Solenoid valve Q0.1 can be activated via a TP timer for a set time.
- When the process signal disappears, a TOF timer can keep an output active for an additional time.
- Pressing the Stop pushbutton I0.1 stops the process.

Ladder (LAD) Programming Logic

The general structure of the program within TIA Portal V20 is shown below.

Network 1 — System Start and Stop

The Start pushbutton I0.0 is used to begin the process. The Stop pushbutton I0.1 is used to stop it.

The logic can include a seal-in using an auxiliary contact of output Q0.0 itself, allowing the system to keep running after releasing the Start pushbutton.

Output Q0.0 represents the process's main motor.

Network 2 — TON Timer: Activation Delay

Activation of a process condition enables the TON timer. For example, when a part is detected by I0.2, the timer starts counting a programmed time of 5 seconds.

During this period, the timer's output Q remains off. Once the 5 seconds have elapsed, the TON's output Q activates and allows the process to move to the next stage.

Application

- Waiting before starting a motor.
- Delaying a solenoid valve.
- Coordinating two process stages.
- Generating a stabilization time.

Network 3 — TP Timer: Pulse Generation

The TON timer's output can be used to trigger a TP timer. When the TP receives the trigger signal, it generates an output pulse for a defined time.

For example, if PT is set to 3 seconds, the TP's output Q will remain active for that period. The TP's output can be used to control solenoid valve Q0.1.

Application

- The TON is activated.
- The TON finishes timing.
- The TP is activated.

- The TP energizes the solenoid valve for 3 seconds.
- Once the time ends, the solenoid valve switches off automatically.

Network 4 — TOF Timer: Off-Delay

The TOF timer can be used to keep an output active after the main signal has disappeared. For example, while motor Q0.0 is running, a fan or auxiliary system can remain active.

When the motor stops, the TOF keeps the output active for the configured time. After the time ends, the output turns off.

This function is especially useful in systems that require ventilation, cooling, or material evacuation after the main process stops.

Network 5 — Run Control and Signaling

Output Q0.2 can be used as a run indicator. While the system is active, the indicator stays lit.

In the event of an abnormal condition, Q0.3 can be used as a fault indicator.

The signaling logic lets the operator visually know the system's current state.

Combined Application Example

An industrial sequence can use the three timers as follows:

- A part is detected by sensor I0.2.
- A 5-second TON timer is activated.
- After the 5 seconds, a TP timer is activated.
- The TP activates a solenoid valve for 3 seconds.
- Once the main process finishes, a TOF timer keeps an auxiliary system active for 10 seconds.
- Once the TOF time ends, the auxiliary output turns off.

This sequence integrates all three timer types within a single automated process.

Programming Best Practices

To develop reliable industrial programs using timers, it is recommended to:

- Correctly define the I/O Factor before starting programming.
- Use descriptive symbolic names for signals.
- Assign identifiable names to each timer instance.
- Use clearly documented setpoint times.
- Avoid duplicating the same timer instance across different networks without technical justification.
- Document each network with comments.
- Verify timer behavior using PLCSIM.
- Check activation and deactivation conditions.
- Analyze system behavior when the input signal disappears.
- Consider the process's stop and safety conditions.

- Verify that programmed times match the system's real requirements.

Well-documented programming makes fault diagnosis, parameter changes, and system maintenance easier.

Practical Exercise

Design a Ladder program in Siemens TIA Portal V20 that meets the following sequence:

Exercise 1: Start/Stop Control

Implement a start/stop system using:

- Start pushbutton I0.0.
- Stop pushbutton I0.1.
- Motor Q0.0.
- Output seal-in.

Exercise 2: TON Timer

When sensor I0.2 is activated, implement a TON timer with a time of 5 seconds. Once the time elapses, the timer's output must activate the next process stage.

Exercise 3: TP Timer

Use the TON timer's output to trigger a 3-second TP timer. The TP's output must control solenoid valve Q0.1. The solenoid valve must remain active only for the set time.

Exercise 4: TOF Timer

Use a TOF timer to keep an auxiliary output active for 10 seconds after the control signal disappears.

Exercise 5: Simulation

Verify the complete program operation using PLCSIM. During the simulation, check:

- TON timer activation.
- Elapsed time ET.
- Activation of the TON's output Q.
- Pulse generation via TP.
- Automatic deactivation of the TP's output.
- Off-delay via TOF.
- Correct operation of the full sequence.

Conclusion

Understanding and correctly applying the TON, TOF, and TP timers is a fundamental skill for developing industrial automation systems with Siemens PLCs.

The TON timer generates a delay before activating an output, the TOF timer keeps an output active for a period after the command signal disappears, and the TP timer generates pulses of a defined duration.

Integrating these instructions in TIA Portal V20 makes it possible to develop precise, orderly control sequences tailored to the needs of industrial processes. Through proper Ladder logic planning, a suitable definition of input and output signals, and validation through simulation, it is possible to build reliable, maintainable, and scalable programs for real-world industrial automation applications.